

2920/203
OBJECT ORIENTED PROGRAMMING
November 2016
Time: 3 hours



**THE KENYA NATIONAL EXAMINATIONS COUNCIL
DIPLOMA IN INFORMATION TECHNOLOGY**

MODULE II

OBJECT ORIENTED PROGRAMMING

3 hours

INSTRUCTIONS TO CANDIDATES

You should have the following for this examination:

Answer booklet.

*Answer **FIVE** of the following **EIGHT** questions.*

All questions carry equal marks.

Candidates should answer the questions in English.

This paper consists of 4 printed pages.

Candidates should check the question paper to ascertain that all the pages are printed as indicated and that no questions are missing.

© 2016 The Kenya National Examinations Council.

Turn over

1. (a) (i) Explain the way in which object oriented programming supports each of the following reasoning:
 - I. induction; (2 marks)
 - II. deductive. (2 marks)
 (ii) Outline **three** typical steps in object oriented programming. (3 marks)
 - (b) Distinguish between *selector* and *setter* operations as used in OOP. (4 marks)
 - (c) (i) Define the term data abstraction as used in OOP. (1 mark)
 - (ii) Peter, a module II student, is convicted that *struct* is an abstract data type. Explain **four** reasons to justify Peter's position. (8 marks)
2. (a) (i) Outline **four** reasons why OOP is popular in software development. (4 marks)
 - (ii) Describe *dynamic binding* as used in OOP. (2 marks)
 - (b) Distinguish between *state* and *behaviour* of an object as used in OOP. (4 marks)
 - (c) Study the following C++ program segment and answer the questions that follow.


```
class point {
    private:
        int x,y;
        point (int i, int j){x=i; y=j;};
    public {
        float midx();
        void displayxy();
    }
};
```

 (i) Identify two errors in the program providing a possible solution in each case. (4 marks)
 - (ii) Explain the implication of using each of the *access specifiers* in the program. (4 marks)
 - (iii) Interpret the 4th line in the program segment. (2 marks)
3. (a) Evaluate each of the following C++ statements (show your working).
 - (i) $b = 12 + 4 - 7 * 9 \% 6 + 12/3$ (2 marks)
 - (ii) $13 - (8\%5) > -9\%7 - 3 + 6 * 2$ (2 marks)
 - (b) (i) Write a C++ program that accepts the radius of a circle and uses an inline function to determine the area of circle. The program should output the area of the circle. (4 marks)
 - (ii) Describe the circumstance under which the following varieties of classes are used in OOP:
 - I. data managers; (2 marks)
 - II. data source/sink. (2 marks)

- (c) Write a C++ program that accepts the dimensions of a rectangle and a rhombus. The program should then use an *overloaded function* named *area* to determine area of the respective figures. Output the dimensions and areas of the figures. (8 marks)
4. (a) (i) Explain the term recursive function as used in programming. (2 marks)
- (ii) Distinguish between *value* and *variable* parameters as used with functions during C++ program writing. (4 marks)
- (b) (i) List **four** rules that are applicable when using constructors in C++ programs. (4 marks)
- (ii) Write a C++ program that will implement a class named *arithmetic* with three integers initialized to 5, 7 and 10 respectively and a member function used to determine and output the product of the integers. Use a parameterized constructor. (5 marks)
- (c) Interpret the following C++ program segment. (5 marks)
- ```
#include < iostream >
using namespace std;
class myclass {
 int a;
public:
 myclass() { a=120; }
 int get();
};
int myclass::get() { return a; }
int main() {
 myclass ob;
 myclass *p;
 p=&ob;
 cout<< "output value using pointer: " << p->get();
 return 0; }
```
5. (a) (i) State **two** ways of using references in C++ programs. (2 marks)
- (ii) Distinguish between *friend function* and *function prototype* as used in C++ programs. (4 marks)
- (b) Explain **three** benefits of *inheritance* as applied in OOP. (6 marks)
- (c) (i) Helen would like to implement operator overloading in her system project. Outline **two** restrictions associated with operator overloading that she should consider. (2 marks)
- (ii) Write a C++ program that will implement a class named *matrix* with following properties: (6 marks)
- data members as x and y;
  - a constructor for initializing the matrix elements (x = 3 and y = 4);
  - a member function for displaying the matrix elements;
  - an overloaded operator function for determining the product of the matrix with a scalar that is input in the main program.
- Note: The program should display the matrix elements of the product.

6. (a) (i) Explain the following forms of inheritance as used in OOP:
- I. extension; (2 marks)
  - II. limitation. (2 marks)
- (ii) Describe the circumstance under which each of the following concepts are most applicable in OOP:
- I. abstract class; (2 marks)
  - II. virtual function. (2 marks)
- (b) Copy constructors are only used for initialization in OOP. Describe **three** occurrences of initialization where they are applicable. (6 marks)
- (c) Write a C++ program that will write characters to a file until the user enters '\$' to stop. (6 marks)
7. (a) Outline **four** disadvantages of implementing inheritance in C++ programs. (4 marks)
- (b) Distinguish between coupling and cohesion as used in OOP. (4 marks)
- (c) Explain the term *destructor* as used in C++ programs. (2 marks)
- (d) Write a C++ program that will carry out the following:
- define a class named *circular* that has data members named *radius* and *height*, a member function named *set* which is used to initialize the value of *length* and *height* and a virtual function named *volume*;
  - implements a derived class from *circular* named *cylinder* whose radius and height are 14 cm and 10 cm respectively;
  - implements a derived class from *circular* named *cone* whose radius and height are 14cm and 5 cm respectively;
  - outputs the volume for the *cylinder* and *cone*.
- Use *pointers*. (10 marks)
8. (a) Explain **two** reasons for using I/O files in OOP. (4 marks)
- (b) Distinguish between overloading unary operators using member functions and using friend functions. (4 marks)
- (c) The module III candidates have been advised to use OOP in their projects. Explain **two** measures that they should put in place in order to cope with the challenges in OOP. (4 marks)
- (d) Write a C++ program that uses an array of objects to generate the following output. Use for loop. (8 marks)
- |   |   |
|---|---|
| 1 | 2 |
| 3 | 4 |
| 5 | 6 |
| 7 | 8 |

**THIS IS THE LAST PRINTED PAGE.**